

CIS 515 Project Proposal Submission- Team 011

Real-Time Wait Time Monitoring for Campus Eateries -

Problem/Opportunity Definition: Students often experience long and unpredictable wait times at popular on-campus coffee shops and eateries, especially during breaks between classes. These delays lead to missed class time, frustration, and inefficient use of campus dining resources. The unpredictability of queues makes it difficult for students to effectively use short breaks, causing unnecessary anxiety and wasted time. Without visibility into the situation at dining locations, students may arrive only to leave without ordering due to long lines, resulting in lost sales for vendors as well.

Why is this Important to Solve: Having real-time visibility into queue lengths can help students plan their visits more effectively, reduce peak-hour congestion, and enable dining services to improve operational planning. It also enhances student experience by saving time and reducing stress. A transparent, data-driven approach can help balance traffic across multiple vendors and minimize overcrowding. Moreover, it fosters a sense of control for students over their daily schedules, contributing to improved campus life and well-being.

Stakeholders and Beneficiaries: ASU students, campus visitors, dining services management, retail vendors (e.g., Starbucks, Chick-fil-A, Pitchforks), and app developers managing ASU's digital platforms. Faculty and staff who also rely on these eateries will benefit from reduced congestion and more predictable service. In the long term, ASU administration could leverage this data to optimize space planning and vendor performance evaluations.

How is it Being Solved Today: Most students check in person, rely on word of mouth, which is not updated/real-time. There is no system-wide digital tracking of queue lengths for ASU eateries. Current strategies are manual, inefficient, and lack any form of predictive or analytical capability. This results in inconsistent experiences and operational blind spots for vendors and campus services alike.

Proposed End-to-End Solution Idea: Install overhead or wall-mounted cameras facing customer queues at key locations. Capture frames periodically and run them through a computer vision object detection model to count the number of people in line. Estimate average wait times using past transaction speed metrics (e.g., people per minute). Feed this data to a backend system that updates live wait times in a dashboard, the ASU app, or nearby digital signage. The system can be designed to anonymize all visual data, focusing only on body count rather than identities, ensuring privacy while delivering real-time insights.

Possible Risks/Roadblocks: Challenges include ensuring accuracy in variable lighting conditions and crowded environments, dealing with privacy concerns, and obtaining permission to install cameras in commercial vendor spaces. Additional complexity lies in integrating with ASU apps or display systems. Technical risks may also involve calibration across different vendor layouts and adapting the model to account for irregular or dynamic queue formations. Furthermore, data security protocols must be strictly followed to prevent any misuse or unauthorized access to visual data streams.